

Networking & Request Flow

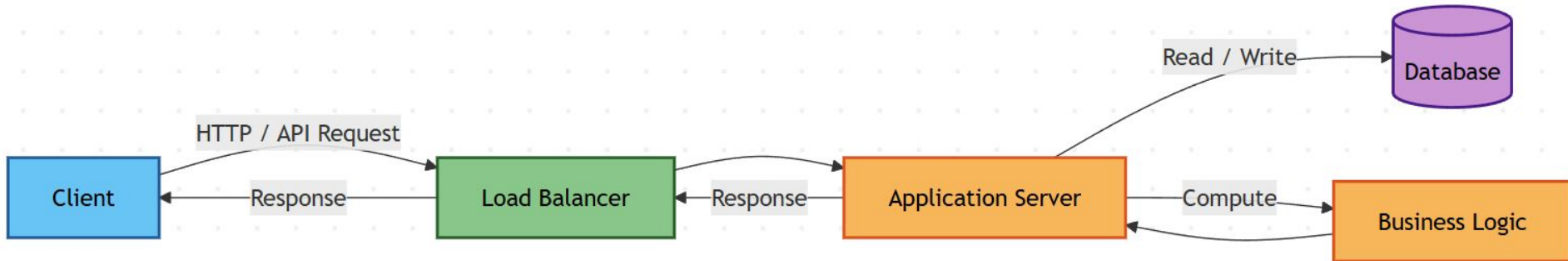
System Design Crash Course: Quick Revision Before Interviews

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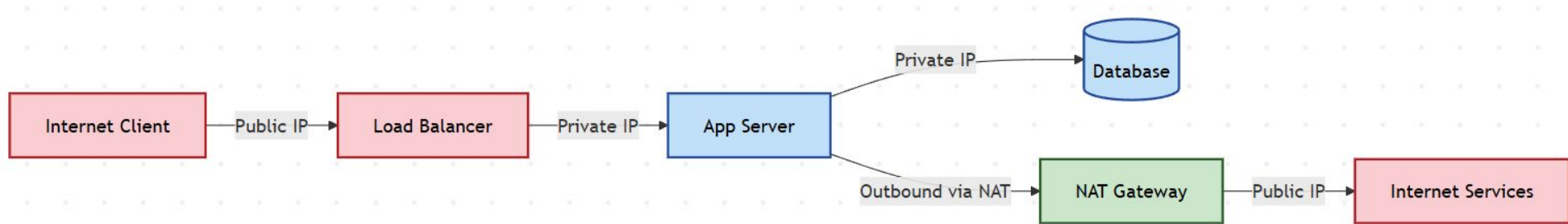
Client-Server Request Flow

1. Client initiates request
2. Server validates and processes
3. Data access and computation
4. Response returned to client
5. Foundation of distributed systems



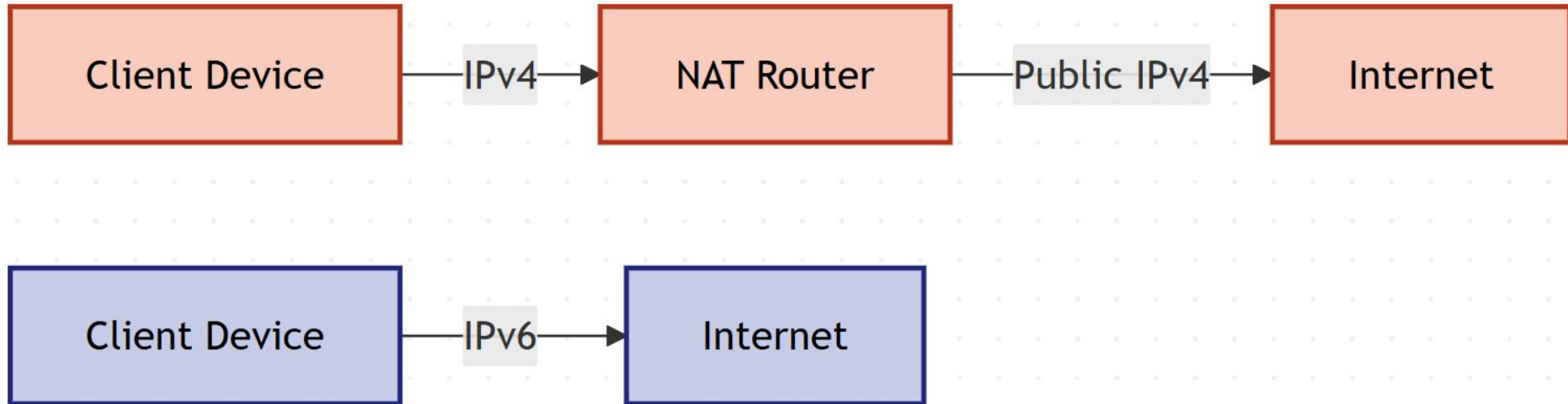
Public vs Private IP Addresses

- Public IP: internet-routable address
- Private IP: internal network address
- NAT maps private to public
- Reduces IP usage, improves security
- Common cloud default pattern



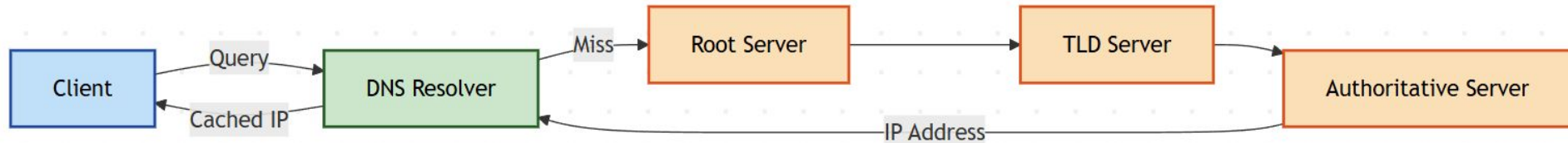
IPv4 vs IPv6

- IPv4: 32-bit address space
- IPv6: 128-bit address space
- IPv4 exhaustion drove NAT
- IPv6 enables end-to-end connectivity



DNS Resolution Flow

1. Client queries DNS resolver
2. Resolver checks local cache
3. Root → TLD → authoritative lookup
4. IP returned and cached
5. Impacts latency and availability



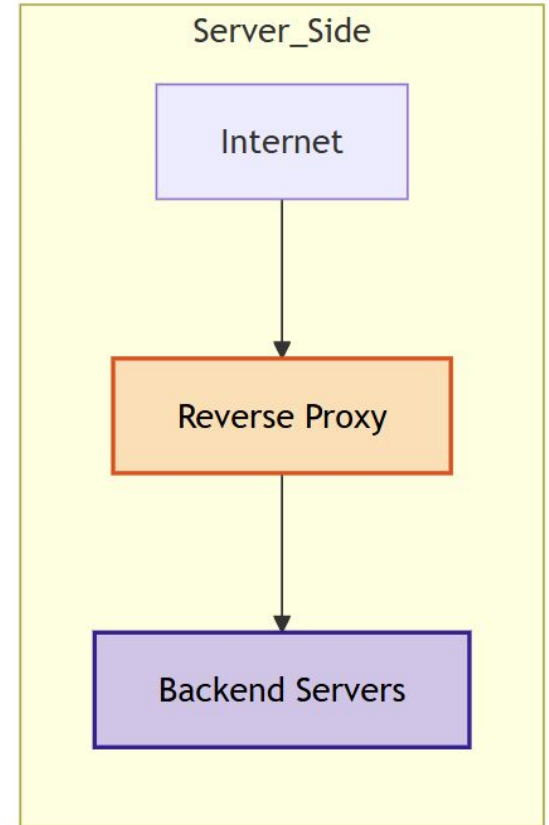
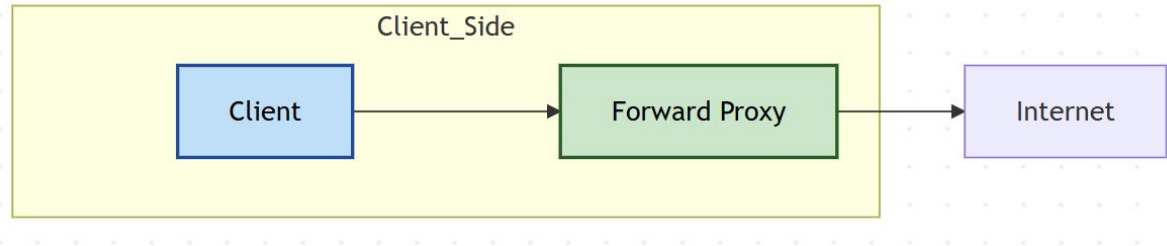
DNS Caching and TTL

- DNS responses are cached
- Caches exist at multiple layers
- TTL controls cache duration
- Longer TTL improves performance
- Shorter TTL improves freshness



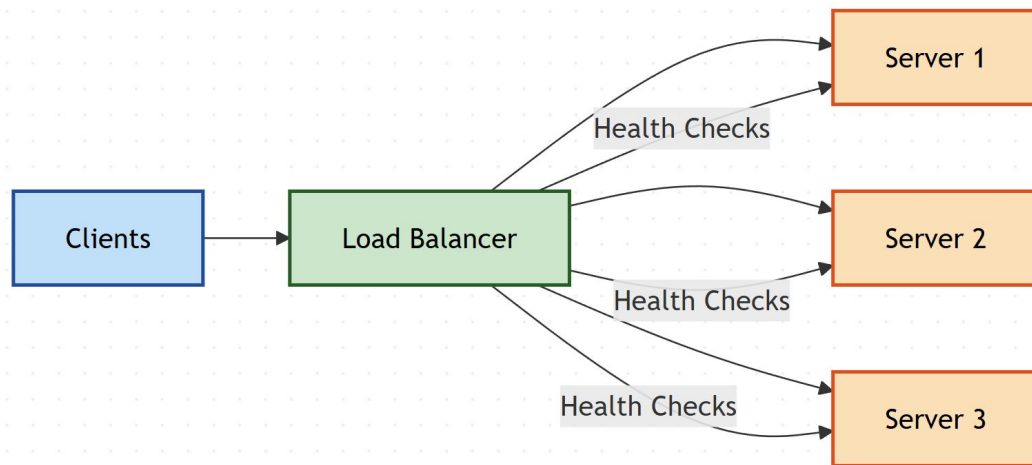
Forward Proxy vs Reverse Proxy

- Forward proxy sits before clients
- Reverse proxy sits before servers
- Forward hides client identity
- Reverse hides backend topology
- Security, scale, control trade-offs



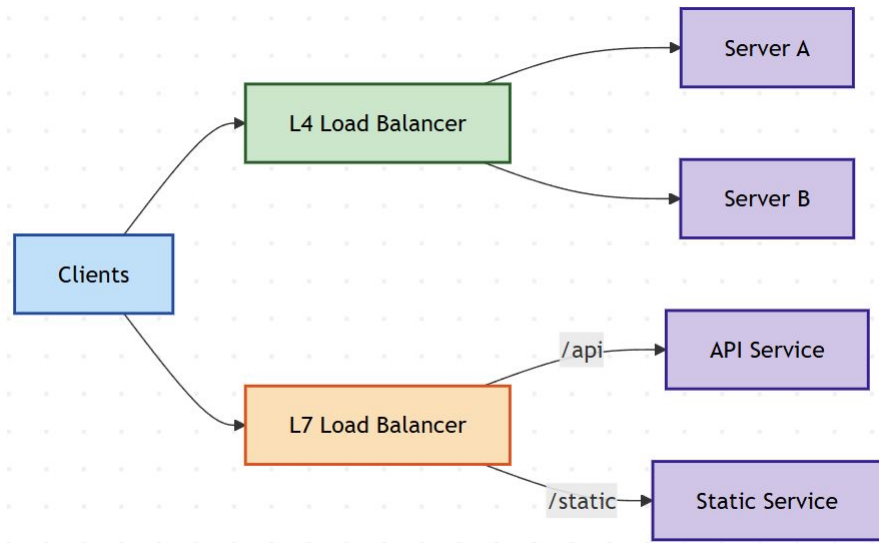
Load Balancer Fundamentals

- Distributes traffic across servers
- Improves availability and throughput
- Uses health checks
- Supports scaling and failover
- Adds latency and complexity



Layer 4 vs Layer 7 Load Balancing

- Layer 4: transport-level routing
- Layer 7: application-level routing
- L4 routes by IP and port
- L7 routes by request content
- Speed versus intelligence trade-off



Content Delivery Networks (CDN)

- Caches content at edge locations
- Serves users from nearest region
- Reduces latency and origin load
- Best for static and media content
- Adds cost and cache complexity

